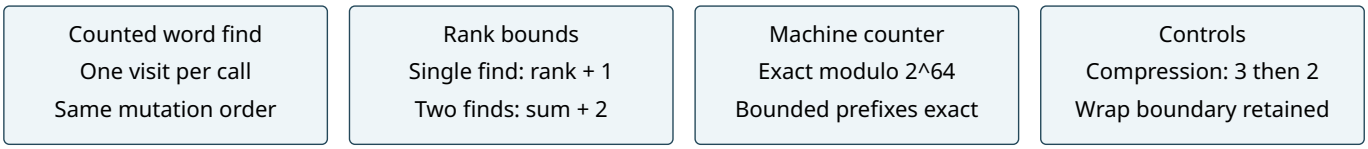
Find visits and exact machine counter accounting

Producer T-1348; archive T-1350. SignedGraph::find at f123ae7053 increments operations before testing whether the vertex is a root. The counted relation retains the prior word-level Run semantics, including reading the potential after the recursive call, and adds one visit for each call.



Result. Eleven named declarations compile. Erasing the count yields the existing Run, and every existing Run admits a counted derivation. Counts are positive. A forest-rank certificate bounds a find from x by rank(x)+1 visits. Using forest preservation between the calls, two finds have at least two visits and at most rank(u)+rank(v)+2. This is a mathematical bound in the supplied rank certificate, not a measured runtime or a tight graph-size bound.

Counter connection. Repeated UInt64 increments equal (initial counter + visits) modulo 2^64. If the initial counter plus the visit bound is below 2^64, each prefix equals the ordinary integer count. Sequential counters compose exactly; for two counted calls under the bound, the final counter is initial+a+b. The construction accounts for the source operations counter only, not field-operation counters or wall time.

Checks. Seven controls establish one root visit, three visits along a two-edge path, two visits on the same vertex after compression, an ordinary single increment, a paired increment, the last non-wrapping increment and the modular wrap boundary. Eighteen named axiom readbacks contain only propext and Quot.sound, or no axioms. Empty, renamed and sorryAx inventories are rejected. A deliberately false claim equating the wrapped word to the mathematical integer 2^64 is rejected.

Assumptions and boundary. Rank bounds use the stated forest invariant. Two-find composition also uses canonical word potentials, a positive modulus at most 2^63 and valid represented equations to apply the existing refinement and forest-preservation result. Exact machine counts require the explicit no-overflow bound. The wrap control describes Lean UInt64 modular arithmetic; it does not assert a particular Rust build's behavior beyond that bound. Stack limits, compiler refinement, other counters and allocation/partial failures remain outside this model. Independent source-fidelity review is pending.

Evidence. Each module check used one thread, 4096 MiB and a 120-second limit. Imported compiled dependencies are reused and hash-bound; no independent kernel/binary recheck occurred. Pre/post hashes do not exclude transient concurrent edits. The first counted-induction attempt and the first counter-sequence proof failed and are retained alongside the successful attempts and deliberate false theorem. The diagram describes proof components, not benchmark measurements.

Next construction. Connect cycles, negative_cycles and zero_coefficient increments to the source dispatcher, preserving their branch conditions and overflow bounds. Then combine counter and component-cardinality contracts with native storage representation. No scientific run, new cost ratio, cohort change or budget renewal occurred.